

Customer Churn Analysis & Prediction

Telecom Industry | 7,032 Customer Records | Python · SQL · Machine Learning

26.6%

Current Churn Rate

5–10%

Industry Benchmark

78.7%

Model Accuracy

Key Findings

Three factors drive the majority of churn

Contract Type

42.7%

Month-to-month churn rate

3× higher than 2-year contracts (2.8%)

Customer Tenure

47.7%

New customer churn (0–12 mo)

Loyal customers (49–72 mo) churn
only 9.5%

Payment Method

45.3%

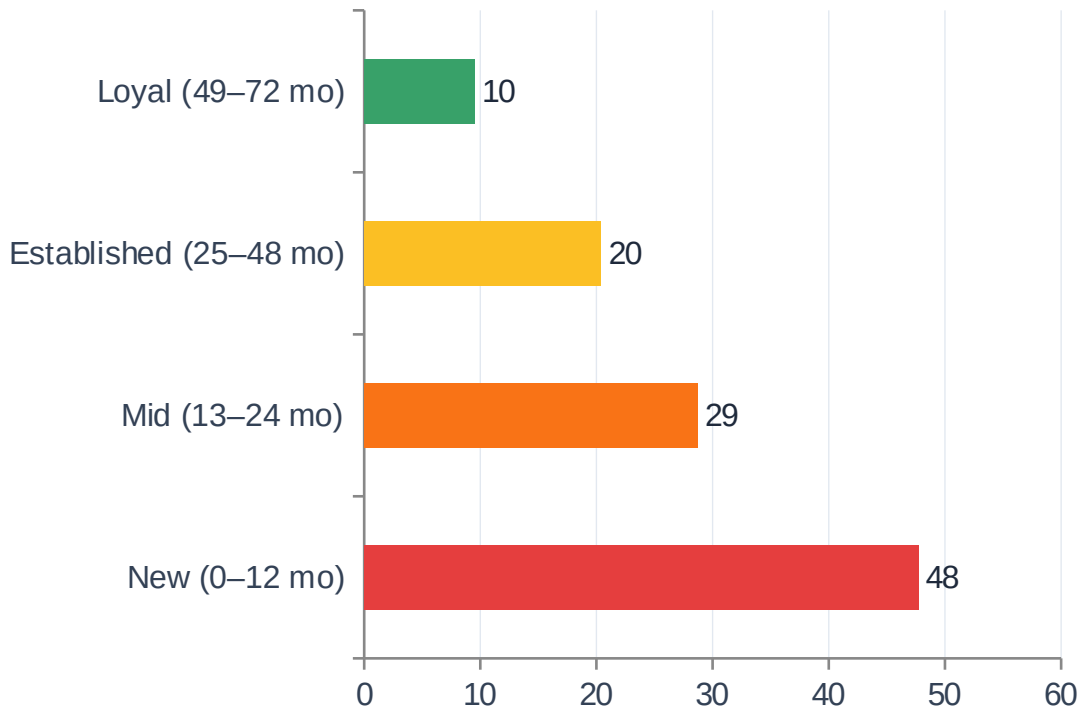
Electronic check churn rate

3× higher than auto-pay customers
(15–16%)

Baseline churn rate: 26.6% | Industry benchmark: 5–10%

Customer Segments at Risk

Churn rate by tenure band (SQL analysis on 7,032 records)



Key Insight

New customers churn at **47.7%**
— 5× higher than loyal customers
at **9.5%**

The first 12 months is the most critical retention window.

ML Model Results

Logistic Regression trained on 5,625 customers • tested on 1,407 unseen customers

78.7%

Overall Accuracy

62%

Churn Precision

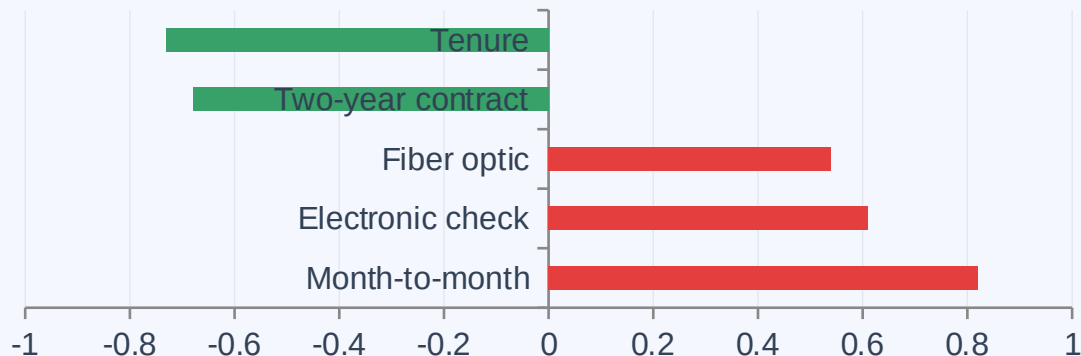
52%

Churn Recall

193

Churners Caught

Top Predictors of Churn



Recall vs Accuracy

We prioritise Recall over Accuracy — missing a churner costs more than a false alarm. 181 at-risk customers were missed and represent significant revenue at risk.

Recommendations

Three targeted actions to reduce churn immediately

01

Early Onboarding Programme

New customers (0–12 months) churn at 47.7%. Introduce a welcome call, onboarding discount, or check-in at month 3 to push customers past the danger zone.

Target: reduce new customer churn from 47.7% → 30%

02

Contract Upgrade Campaign

Month-to-month customers churn at 42.7% vs 2.8% on 2-year plans. Offer incentives (price lock, free add-ons) to convert monthly customers to annual contracts.

Target: convert 20% of monthly customers to annual plans

03

Auto-Pay Migration Drive

Electronic check customers churn at 45.3% vs 15–16% for auto-pay. Offer a small monthly discount for switching to automatic payment methods.

Target: reduce electronic check churn from 45.3% → 25%